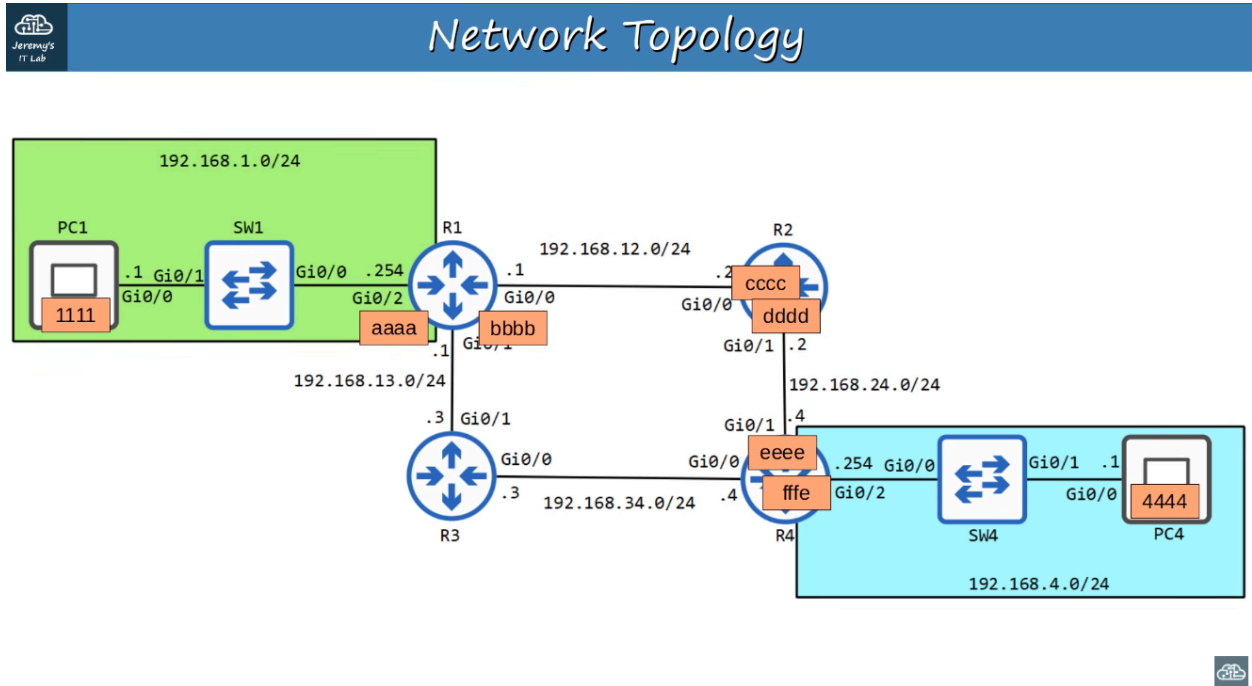


Day 12: The Life of a Packet

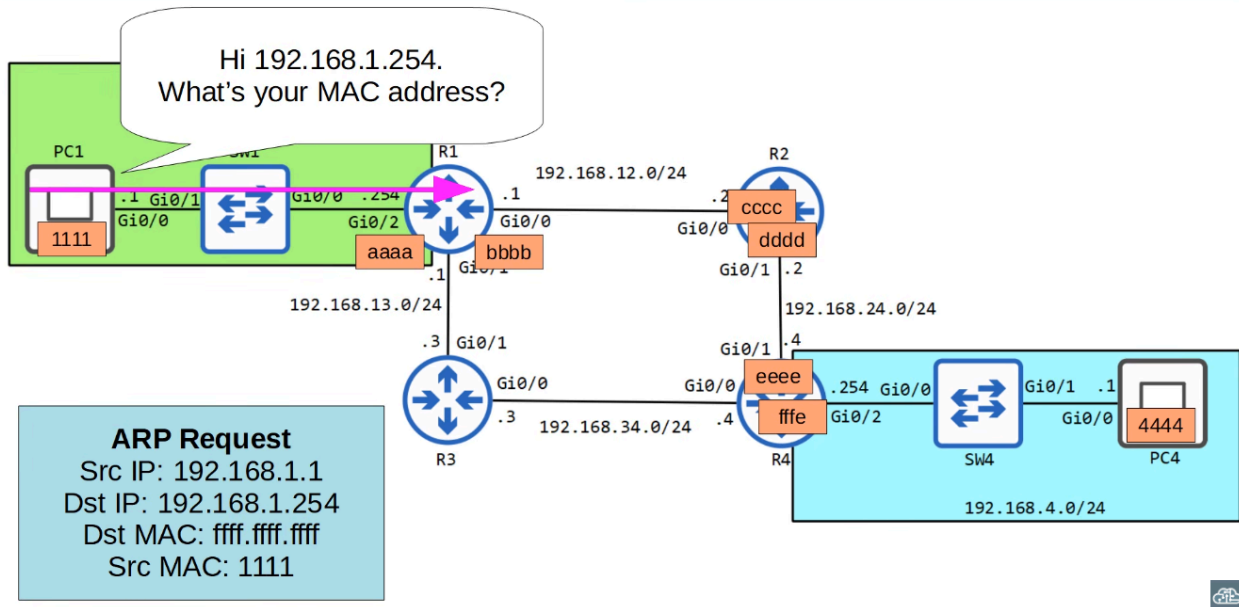


EACH Network device's interfaces have a UNIQUE MAC Addresses.

PC1 to R1

PC1 has not sent any packet before, so it must use ARP to get others MAC Address first.

ARP (Address Resolution Protocol)



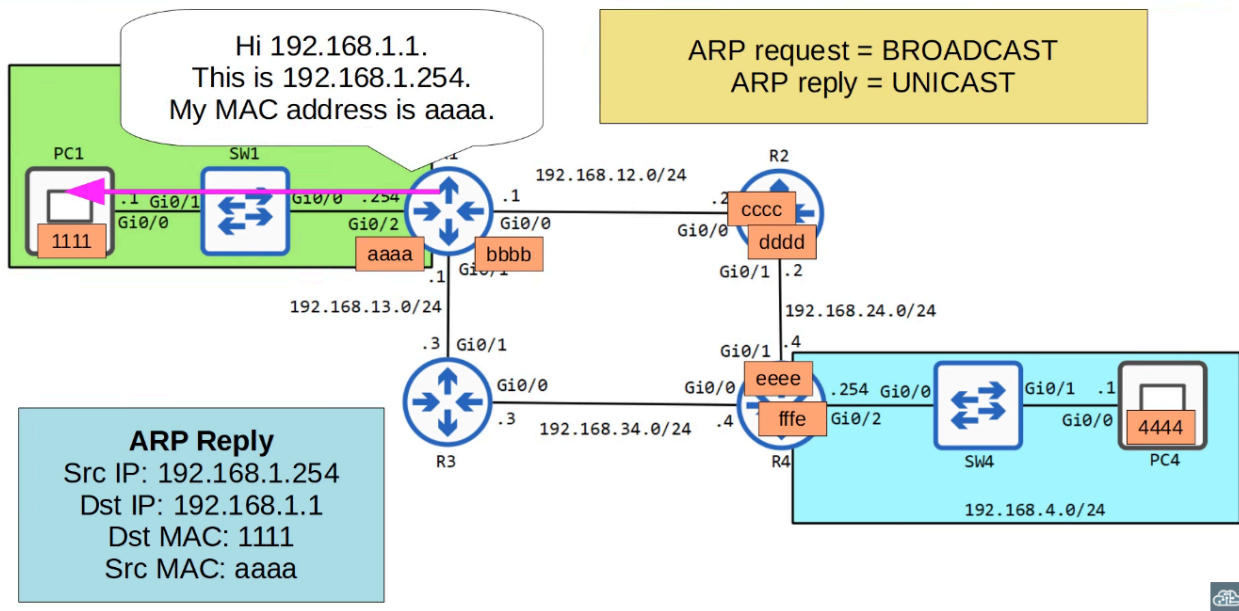
In a TCP HEADER:

- SOURCE IP ADDRESS comes before DESTINATION IP ADDRESS

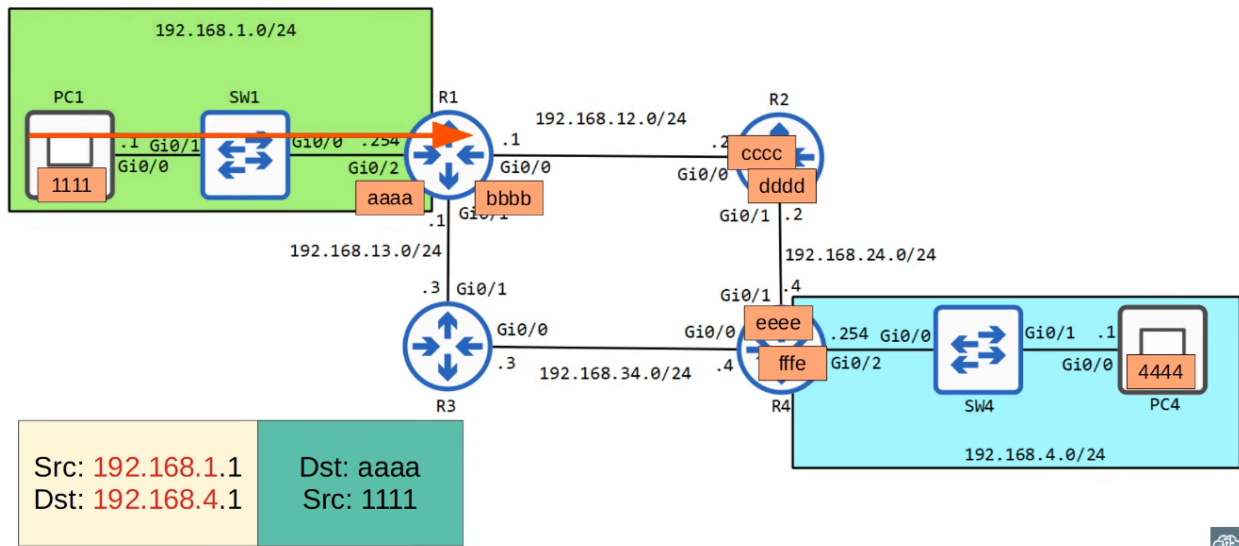
In an ETHERNET HEADER:

- DESTINATION MAC ADDRESS comes before SOURCE MAC ADDRESS

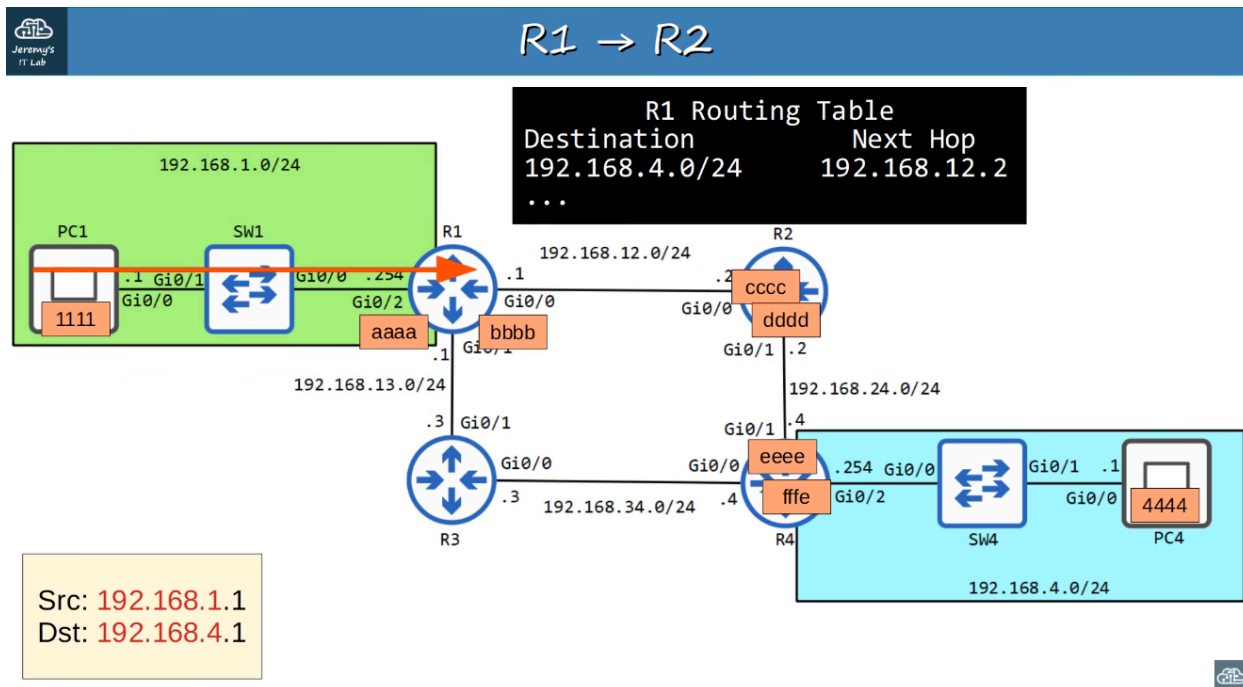
ARP (Address Resolution Protocol)



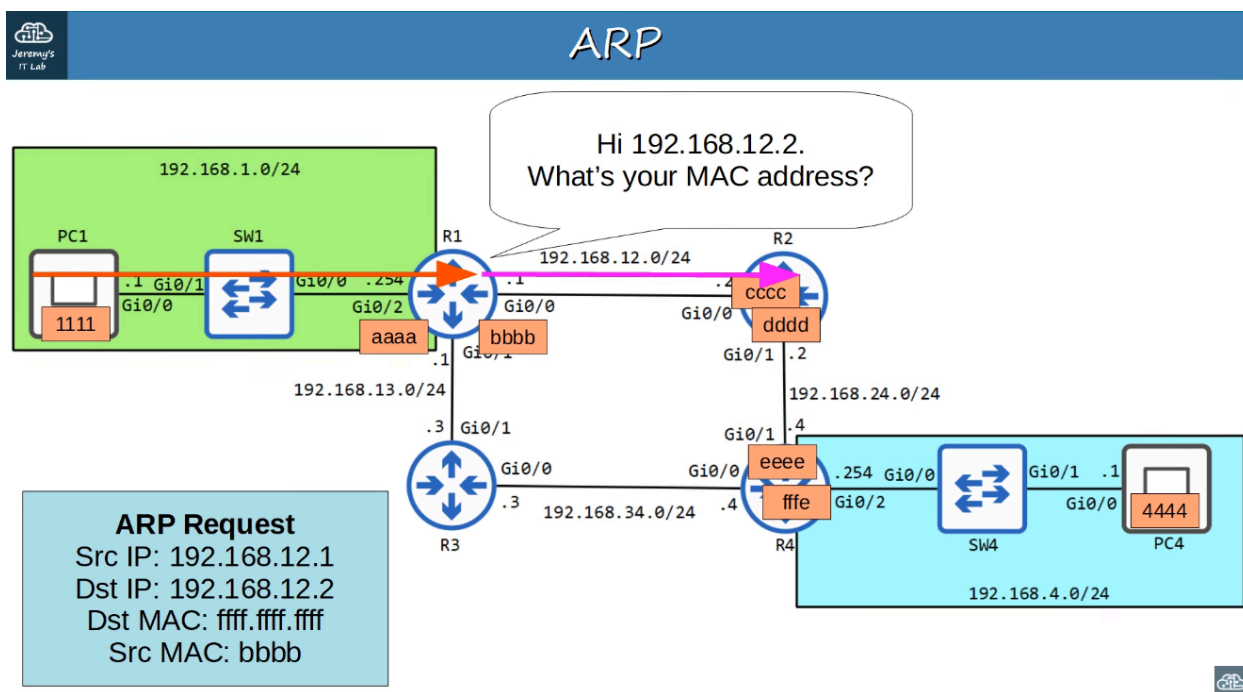
PC1 → R1



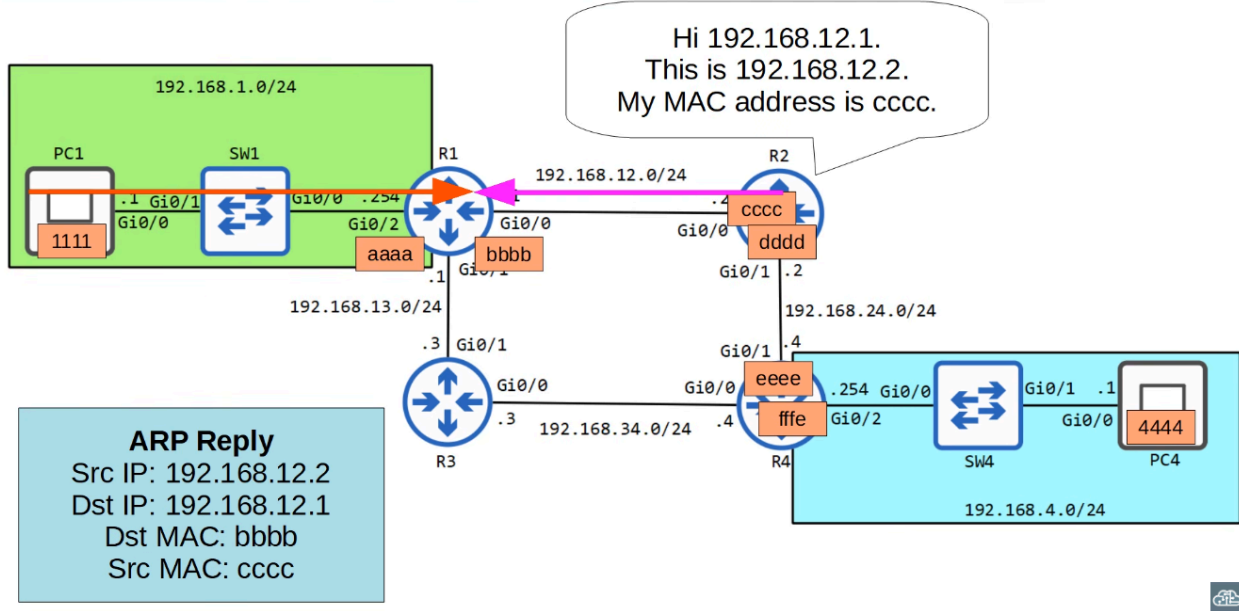
R1 to R2



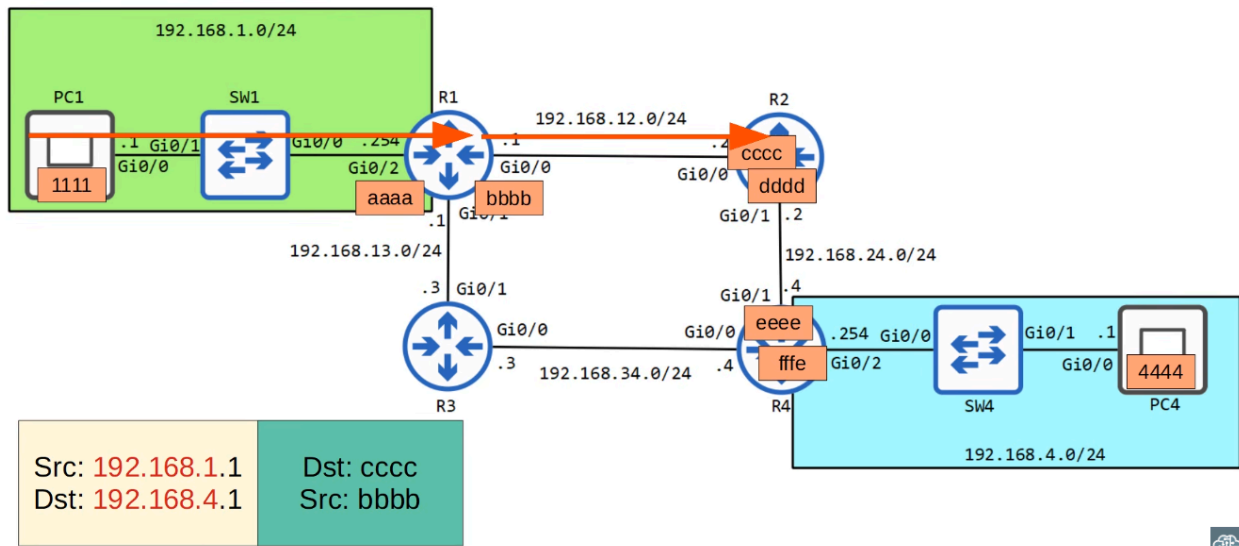
Before R1 sends packet to R2, it must learn R2's MAC Address by using ARP



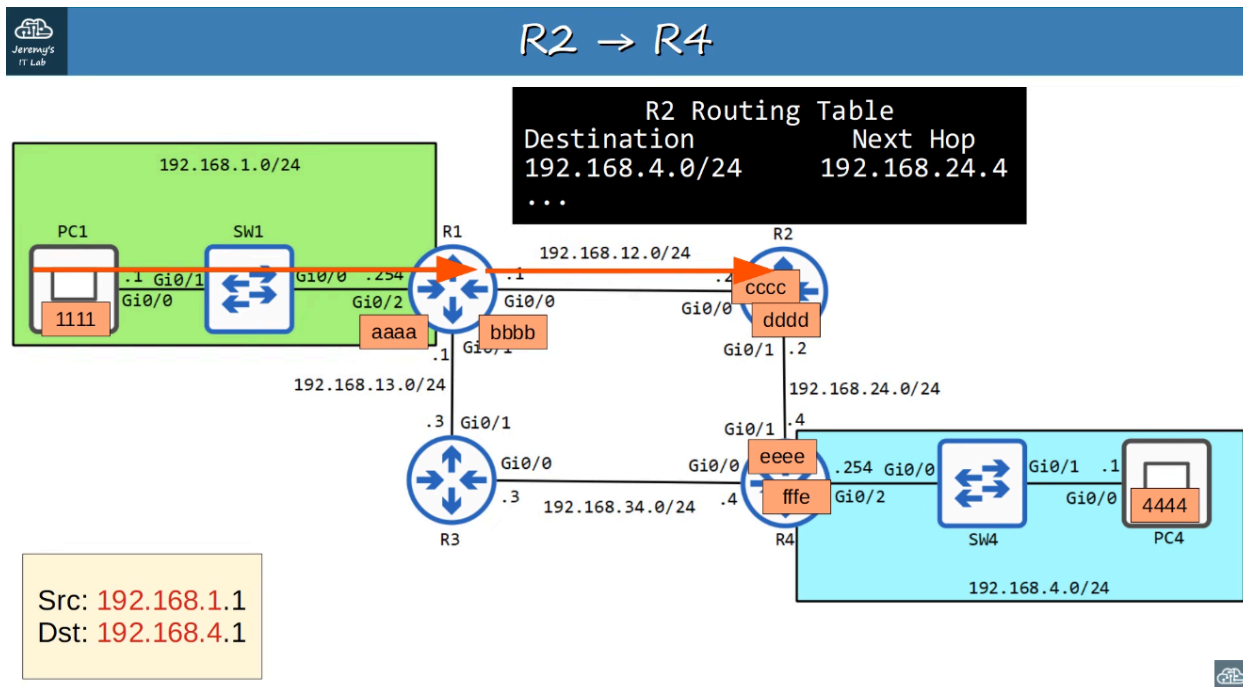
ARP



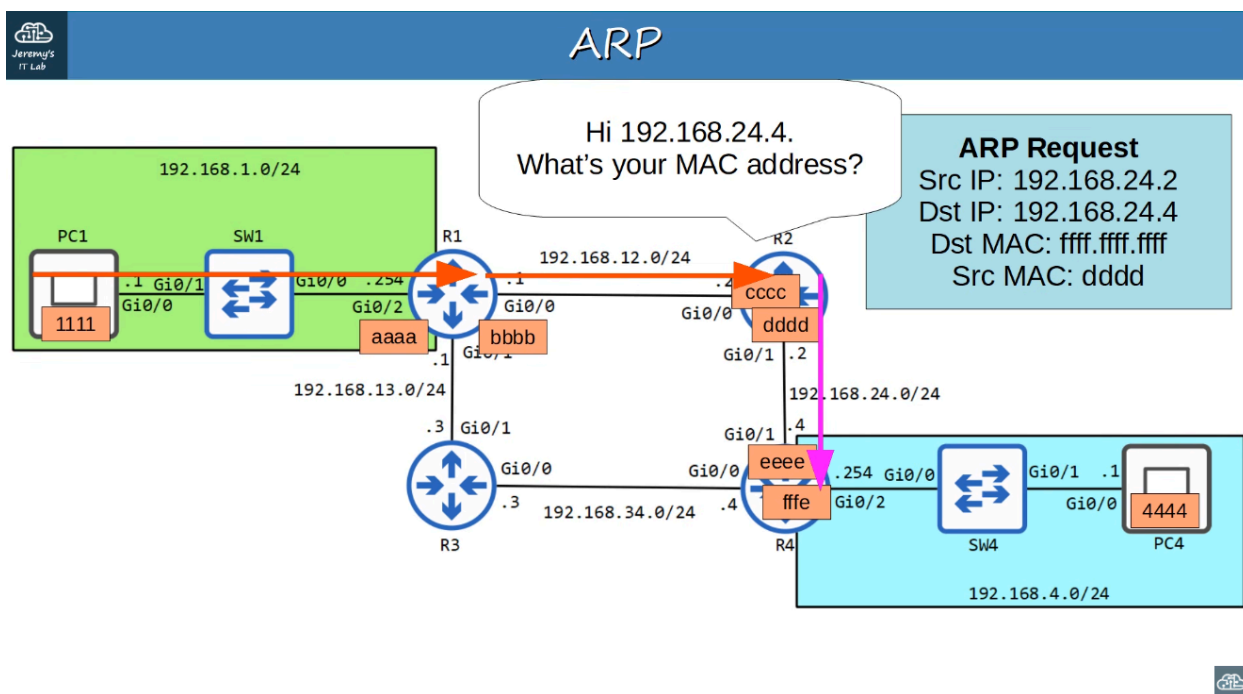
R1 → R2



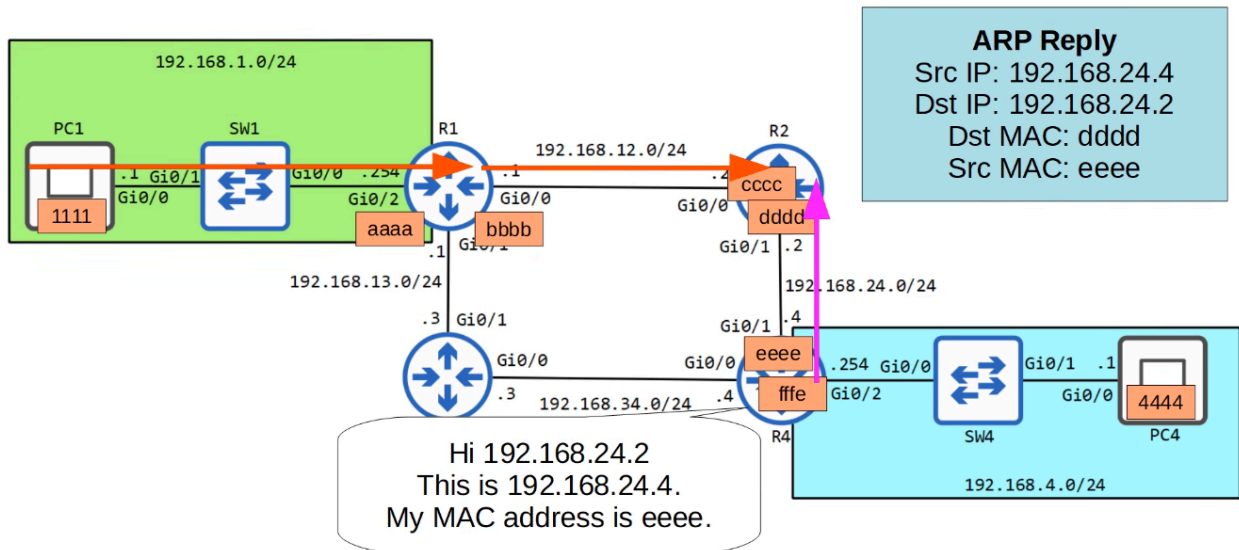
R2 to R4



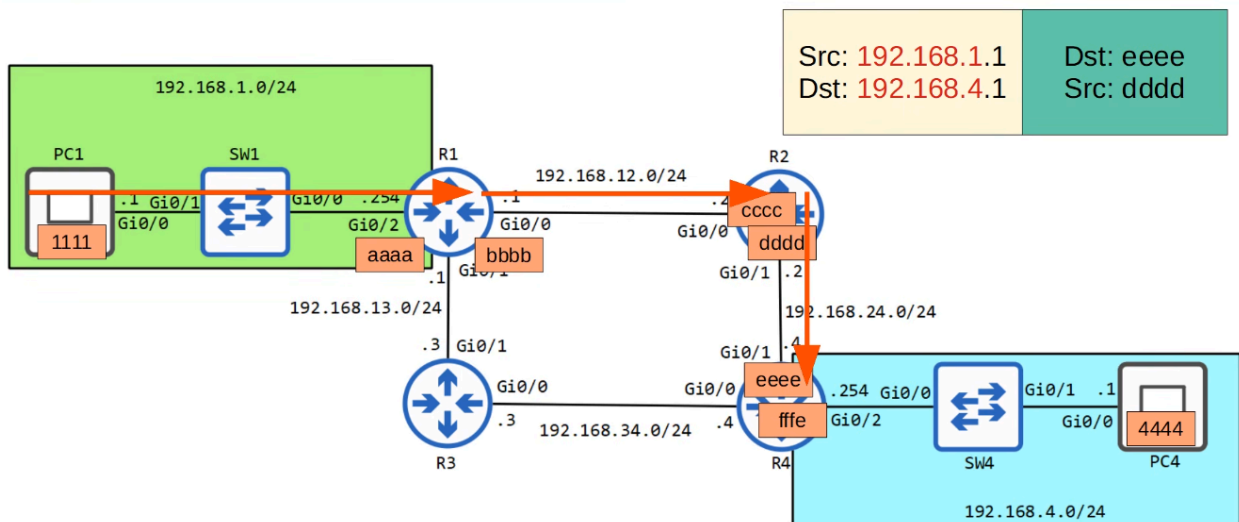
Same as the previous progress, R2 must use ARP to get R4 MAC Address first



ARP

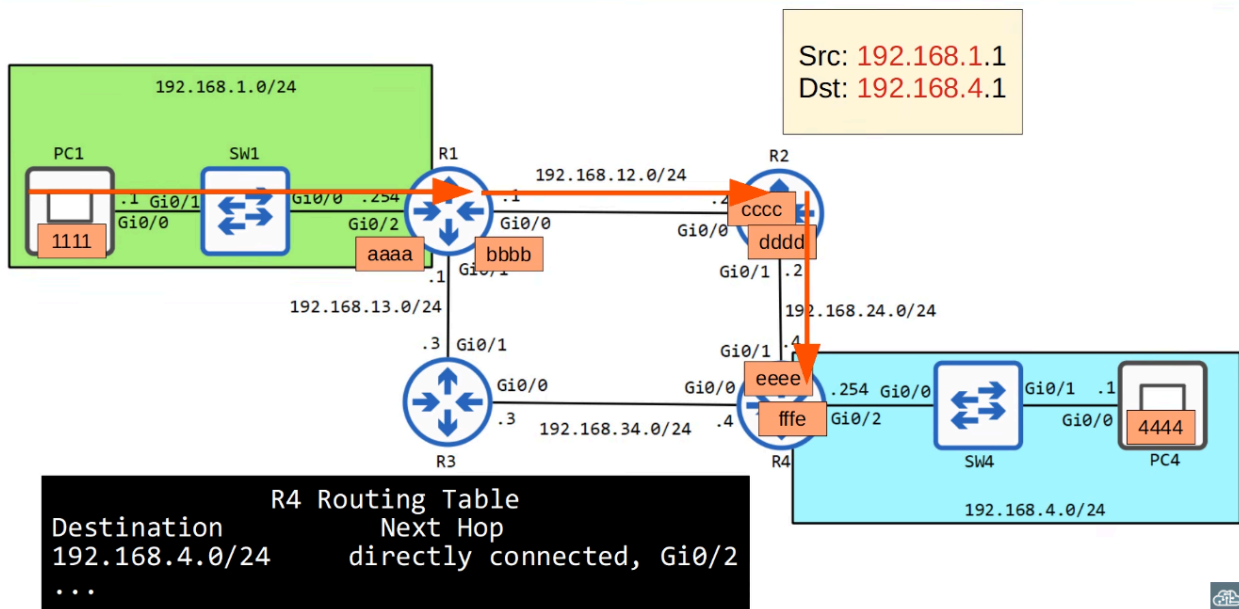


R2 → R4

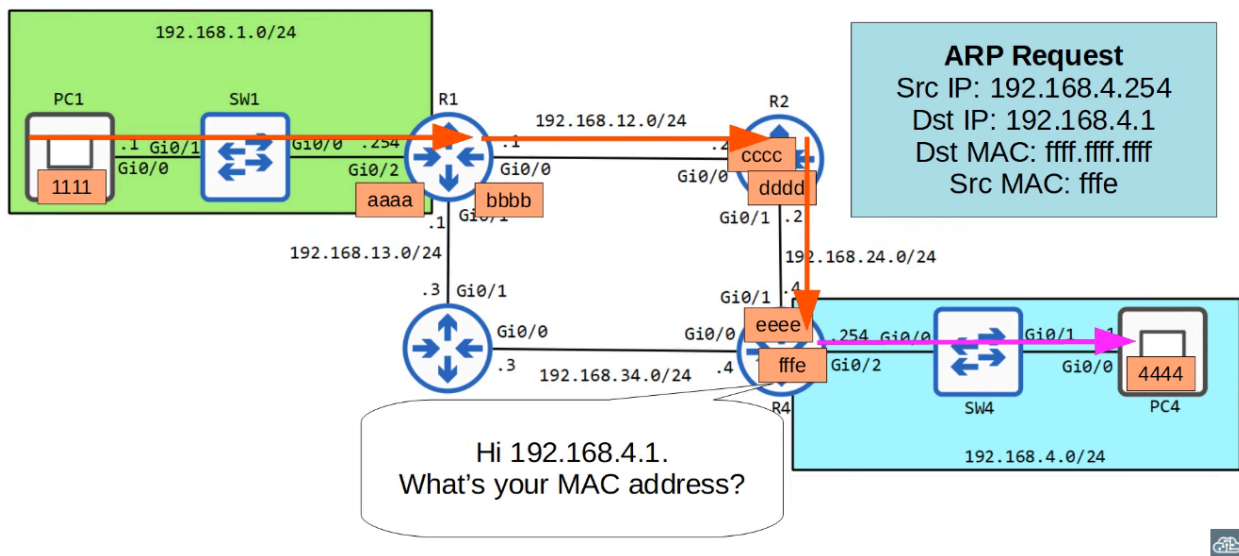


R4 to PC4

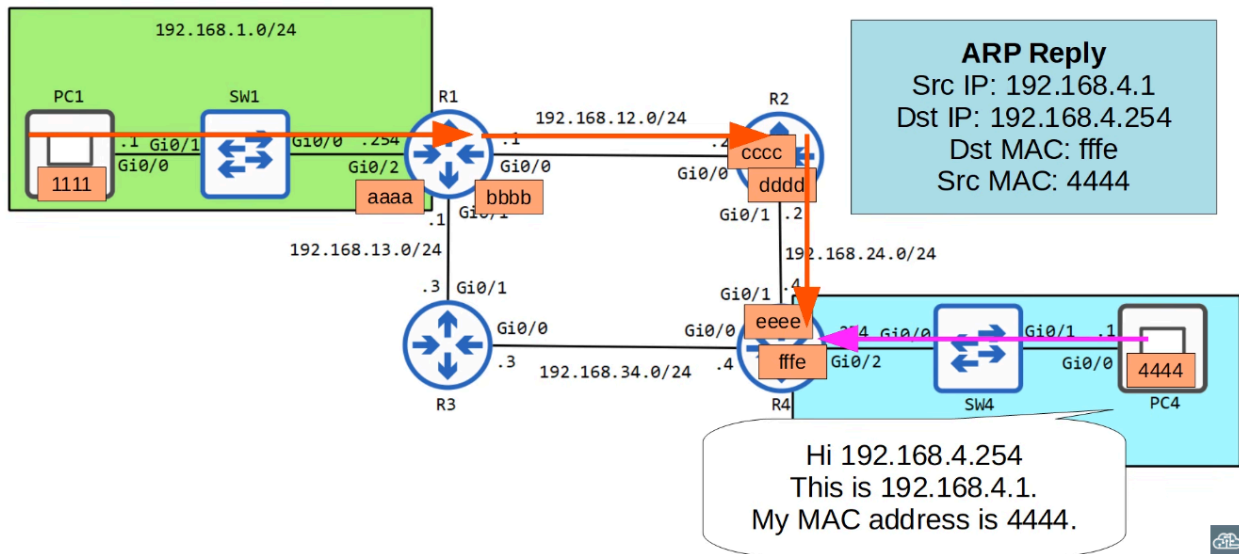
R2 → R4



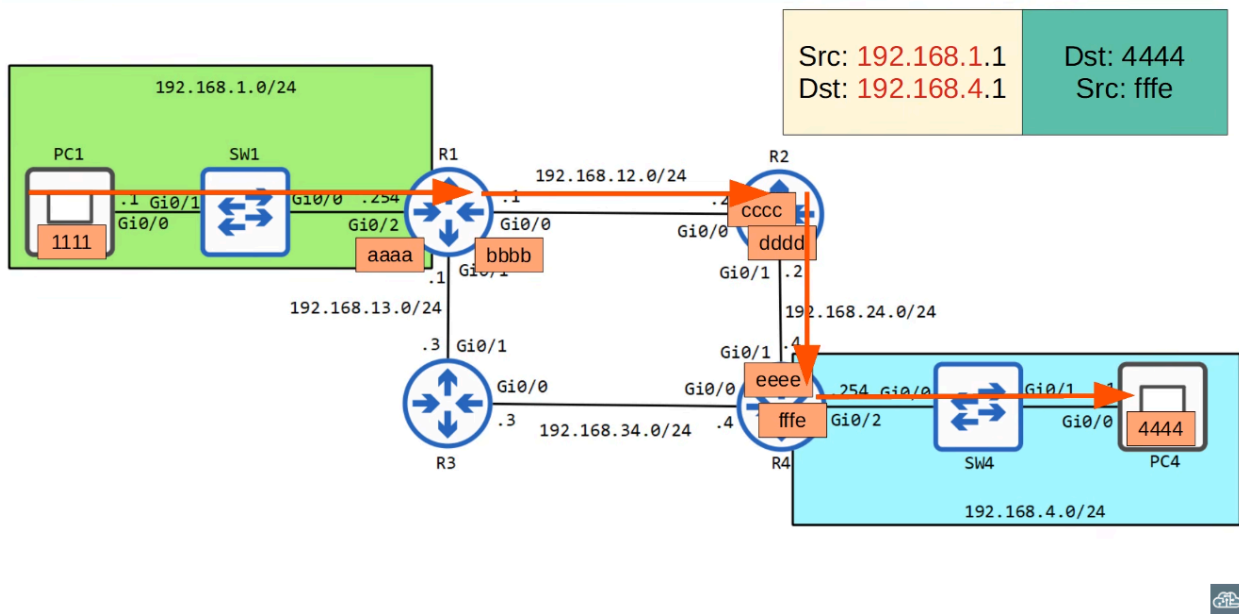
ARP



ARP



R4 → PC4



When a HOST sends a packet to another HOST, the SOURCE or DESTINATION IP doesn't change - even though ROUTERS may change the ETHERNET HEADER (SRC/DEST MAC ADDRESSES).